Empirical Backtesting of the BIS Climate-Extended Vasicek Model: An out-of-sample Credit Risk Study Assessing Agricultural Loan Defaults During the 2022 European Compound hot-dry event

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Abstract

The impact of climate risk in credit lending is often basic or entirely omitted; the increasing frequency and severity of extreme weather events across the globe are straining this simplification, especially in highly affected sectors such as agriculture. With the institutional implementation of credit risk frameworks such as the Vasicek Credit Risk model, the question of the feasibility and efficacy of adjusting these models to incorporate climate risk is becoming an increasingly prevalent issue in risk management. By using non-performing loan (NPL) data to stand as an observable proxy for loan default probability across Spain, Turkey, and Serbia during the 2022 compound hot-dry event, this paper cross-regime tests the Bank for International Settlements (BIS) climate-extended Vasicek model by comparing baseline NPL rates excluding and then including an added climate component ($\hat{a} = 0$ & $\hat{a} > 0$) to the model. Having achieved a mean absolute error (MAE) of 3.5641% and root mean square error (RMSE) of 3.6751% the model has proven to be statistically stable in macroeconomic forecasting while crucially having a low variance, signifying that its predictive errors are uniformly distributed rather than having single extreme outliers for the climate event.

I - Introduction

While the physical impact and extent of climate change differ from nation to nation, the underlying mechanics remain the same. Climate Change is getting quantifiably more frequent, more severe and more often than not costlier to respond to. In Africa over 34% of recorded droughts in the past 26 years have occurred in just the last 9 years and in Europe this figure jumps to 37%.[1] Frequent extreme heat and wildfires putting added strain on fresh water, this water strain and consequent scarcity that can affect up to 70% of the population in Southern Europe during seasonal summer stress.[2] Agriculture in Europe 2025 accounted 50% of total annual European water consumption.[3] This leads to the agricultural sector being on the front lines of the effects of climate change, while these concepts are well understood by the vast majority of credit analysts, accompanying credit risk models that specifically focus on better adapting to

these ever increasingly extreme weather events are far less developed. Associated risks are often treated as amplifiers for pre-established risk categories and while regulatory reporting and requirements are close to being standardised, predictive mathematical risk modelling on climate impacts are still some years away.

As climate related unaccounted risk exposure in many major financial institutions, particularly those dealing directly with agri-finance grows,[4] traditional risk models have to pivot from treating physical climate hazards as a negligible secondary element akin to how air resistance is treated in basic mechanics. This developing physical and financial reality has driven international institutions, such as the BIS to attempt to provide the solution by extending pre-established asymptotic single risk regulatory credit risk frameworks under the Basel II/III accords. The use of the Vasicek credit model as a base bridges the difficulty of standardising a risk that is inherently high geolocalised, something that can be affected by something as minor as high salt content in a crop field to something as general as rainfall. The concept is extending already heavily integrated models by embedding climate risk as a systematic factor in the broader economy that itself affects systemic latent asset returns and thus default rates. This new approach that prioritises structural shifts rather than static risk multipliers or percentage addons after the main output, provides a much more grounded method that mirrors real-life behaviour as well as being a highly practical and viable future mathematical modelling tool.

This paper uses the 2022 European compound hot-dry event as an out-of-sample to backtest and empirically test if the climate-extended Vasicek credit risk model accurately captured the additional climate risk relative to the base Vasicek model. This was achieved by comparing the pre-event baseline NPL ratios across the agricultural sector spanning distinct economic development stages, Spain, Turkey and Serbia from 2021 to early 2022 to the post climate event NPL ratio levels in Q3 2022 to Q3 2023. The report focuses specifically on the “probability of default conditional on y in the case of ‘binary’ climate risk” model as the climate extended version. While there is a comprehensive selection of other models outlined in the paper, this particular model relies on publicly available data inputs rather than more complex microeconomic sector assessment which needs proprietary data inputs for parts of the formula. The extended model demonstrates a strong level of precision with an MAE of 3.5641% and RMSE of 3.6751%, this small difference of 0.111% indicates that the model itself is quite precise. However the precision is still up for debate, while the climate-extended model was able to correctly identify across all 3 countries that incorporating climate lead to a rise in conditional probability of default (PD), regional deviations still occurred showing the lack of localised precision. This predominantly arises from the inability to incorporate specific infrastructure, intervention and legal buffers in each country's agricultural systems into the climate-extended model.

The remainder of this paper is structured as follows: Section 2 outlines the theory and mathematical details of the climate-extended model; Section 3 outlines the data collection and coding methodology used along with individual regime characteristics and their relevance in the processing of the data; Section 4 outlines the results of the backtesting and error analysis as well as an individual breakdown of each countries results and possible reasons for the specific outcomes; Section 5 outlines the conclusion and the possible policy implications of regulatory capital and central bank stress testing in the future.

II - Theoretical Framework

The Climate extended Vasicek credit risk model can be derived from the Merton single-asset return model as follows,

$$R_i = \sqrt{\rho}\ Z + \sqrt{1\ -\ \rho}\ \xi_i$$

Where R_i is the standardised latent asset return of a borrower i, Z a systematic risk factor, ξ_i being borrower specific idiosyncratic risk, ρ representing the correlation between the asset and the systematic risk factor parameter, $\sqrt{\rho}$ the systematic risk sensitivity factor, $\sqrt{1\ -\ \rho}$ the idiosyncratic risk sensitivity. A borrower is in default if the normalised asset returns are below a set critical threshold $R_i < C_i$ found using the unconditional probability of default (P_D^0). As R_i obeys the standard normal distribution (SND), the threshold is defined by the inverse cumulative distribution function (CDF), Φ^{-1} so $C_i = \Phi^{-1}(P_D^0)$. Determining the conditional default probability requires defining the probability of default given a realised economic Z value and as this value is fixed alongside $\sqrt{\rho}$, $\sqrt{\rho} \bullet Z$ becomes a static value in the probability operator. At this point all values other than ξ_i are constant and ξ_i follows the standard normal distribution ($\xi_i \sim N(0,1)$), representing the idiosyncratic risk due to the borrower. To transition from an unconditional model to one that is point-in-time, the future economic climate needs to be dependent on a given value at the time given $P(Z) = P(R_i < C_i \mid Z)$. Now Z is deterministic and fixed:

$$R_i < C_i$$

$$\sqrt{\rho}\ Z + \sqrt{1\ -\ \rho}\ \xi_i < C_i$$

$$\sqrt{\rho}\ Z + \sqrt{1\ -\ \rho}\ \xi_i < \Phi^{-1}(P_D^0)$$

$$\sqrt{1\ -\ \rho}\ \xi_i < \Phi^{-1}(P_D^0) - \sqrt{\rho}\ Z$$

$$\xi_i < \frac{\Phi^{-1}(P_D^0) - \sqrt{\rho}\ Z}{\sqrt{1-\rho}}$$

$$P(Z) = P(\xi_i < \frac{\Phi^{-1}(P_D^0) - \sqrt{\rho}\ Z}{\sqrt{1-\rho}})$$

And as ξ_i is distributed using the standard normal distribution, the probability of falling below a specific value is given by the standard normal cumulative distribution Φ and thus yielding the Vasicek Conditional Default Probability formula;

$$P(Z) = \Phi\left(\frac{\Phi^{-1}(P_D^0) - \sqrt{\rho}\ Z}{\sqrt{1-\rho}}\right)$$

This provides the base for the climate extended version. The specific model used in this paper is the "the probability of default conditional on y in the case of 'binary' climate risk", the model is as listed below:

$$P_D\,(default = 1 \mid S = -y) = (1-q)\Phi\left(\left(\frac{1}{\sqrt{1-\rho}}\left(\sqrt{\rho}\,y\;+\;\Phi^{-1}(P_D^0)\right)\right)+\;q\,\Phi\left(\frac{1}{\sqrt{1-\rho}}\left(\sqrt{\rho}\,y\;+\;\Phi^{-1}(P_D^0)\right)+\frac{\hat{a}}{\sqrt{1-\rho}}\right)\right)$$

The additional variables are: q the hazard's probability of occurring in a given timeframe, (1 - q) the probability that the hazard does not occur, creating the weights for an economic baseline where 100*(1-q)% of the time an event does not occur. The variable â represents the damage intensity indicating the extent of the losses due to the climate event, as a volatility-standardised loss on the asset return ($\hat{a} = \frac{\alpha}{\sigma}$) to shift baseline distance to default in terms of standard deviations.

The BIS binary climate model implements climate risk by weighting the probability of a climate hazard occurring, the 2 outcomes rely on the state of a hazard H, H∈ {0, 1}. When H=0 a hazard does not occur and its probability is 1-q, and if H =1 a hazard occurs and the probability of this event is q. If a climate hazard were to occur the impact would be damage to the asset and thus a negative impact on the asset returns, $R_i \Rightarrow R_i - \hat{a}$, so default will now occur when $R_i - \hat{a} < \Phi^{-1}(P_D^0)$, ~ N(0,1²)

$$\textit{Substituting in Merton's SAR formula: } \sqrt{\rho}\,Z + \sqrt{1\;-\;\rho}\,\xi_i - \hat{a}\;<\;\Phi^{-1}(P_D^0)$$

$$\textit{Isolating for borrower specific idiosyncratic risk: } \xi_i\;<\;\frac{\Phi^{-1}(P_D^0)+\hat{a}-\sqrt{\rho}\,Z}{\sqrt{1-\rho}}$$

As ξ_i ~ N(0,1), applying Φ yields the the probability of default given a climate hazard occurring:

$$P(default|Z,\;S\;=\;1) = \Phi\left(\frac{\Phi^{-1}(P_D^0)+\hat{a}-\sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right)$$

Using the law of total probability for a binary event, $P(A) = P(A|B)P(B) + P(A|B')P(B')$ by combining the $P(default|Z,\;S\;=\;0)$ which is equivalent to the standard Vasicek credit risk model and weighting each outcome branch by its probability of occurrence 1-q & q onto expressions 7 & 8 respectively according to the original BIS paper , the probability of default when factoring climate events and their probabilities is:

$$P(Default|Z)\;=\;(1\;-\;q)\bullet P(Default|Z,\;S\;=\;0)\;+\;q\;\bullet\;P(Default|Z,\;S\;=\;1)$$

$$P(default|Z)\;=\;(1\;-\;q)\Phi\left(\frac{\Phi^{-1}(P_D^0)\;-\;\sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right)+q\Phi\left(\frac{\Phi^{-1}(P_D^0)+\hat{a}-\sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right)$$

To express the probability under macroeconomic stress, we redefine Z as -y. Here a positive value of y represents an economic downturn and thus Z takes on a negative value to signify the unfavourable economic state. Substituting Z = -y, $\left(-\sqrt{\rho}\,Z \rightarrow -\sqrt{\rho}\,(-y) \rightarrow \sqrt{\rho}\,y\right)$, into the probability of default when factoring climate events and their probabilities formula while also factoring out $\frac{1}{\sqrt{1-\rho}}$ gives;

$$P_D(default = 1 | S = -y) =$$

$$(1 - q)\Phi\left(\frac{1}{\sqrt{1-\rho}}\left(\sqrt{\rho}\,y + \Phi^{-1}(P_D^0)\right)\right) + q\Phi\left(\frac{1}{\sqrt{1-\rho}}\left(\sqrt{\rho}\,y + \Phi^{-1}(P_D^0)\right) + \frac{\hat{a}}{\sqrt{1-\rho}}\right) *$$

Core Assumptions

- Conditional on the systemic macroeconomic factor y, individual borrower defaults being statistically independent of one another with all portfolio defaults driven entirely by shared macroeconomic exposure
- Portfolios are infinitely granular and thus in a portfolio of loans idiosyncratic risk is entirely diversified away leaving behind purely systematic risk.
- Climate hazards are entirely exogenous and physical climate shocks and the baseline macroeconomic factor y are strictly orthogonal of one another.
- The macroeconomic state dictates borrower default rates, however localised clusters of defaults do not feed back into the macroeconomy.

Mathematical Boundary Conditions

- As damage intensity and/or the probability of a climate hazard occurring tend to 0, $\hat{a}/q \Rightarrow 0$, the binary climate model collapses back to the standard Basel asymptotic single risk factor model.
- Conditional probability of default at extreme macroeconomic states are rational. As the economy booms ($Y \Rightarrow +\infty$), the probability of default tends to 0 ($P_D \Rightarrow 0$). Conversely the opposite is also true, ($Y \Rightarrow -\infty$) & ($P_D \Rightarrow 1$). Both the conditional probability of default and hazard probability q are strictly in the interval [0, 1].

III - Methodology

The empirical backtest evaluates the climate-extended Vasicek model across 3 countries: Spain, Turkey and Serbia. While the 2022 compound hot dry event is classified as a single pan-continental climate event, its localised physical intensity and effects varied greatly. These countries were chosen to provide a broad range of economic and physical conditions for the model to be tested on. Spain was chosen due to its position at the epicentre of severe direct heat channeled northward from the Sahara, but also due to its position as a relatively rich primary EU member. As well as its economic position it also has a highly advanced and structurally robust public-private agricultural insurance system: the Combined Agricultural Insurance System (CAIS). Additionally Spain is also more exposed to climate related risk than other primary EU members due to its significantly higher agricultural output relative to GDP added, 2.9% agricultural gross agricultural value added (GVA) [5] compared to the next highest primary EU member Italy at 1.8%.[6] Turkey

was chosen due to its position as an upper middle income country [7] providing an economic and structural middle ground. The state supported agri-insurance pool (TARSİM) covering around 23% of total cultivable farmland covered by insurance, in between Spain's 55-60% and Serbia's 10% providing an economic middleground to test the models behaviour in a Newly Emerging Economy (NEE).[8] Turkey has also experienced a strikingly volatile climate in 2022 as widespread drought was occasionally interjected with localised torrential downpours as cold and hot air masses collided. These conditions created large scale crop destruction as violent supercells created hail storms and heavy rains lead to flooding, destroying large volumes of crops and a primary reason for the food related hyperinflation in Turkey. Serbia was chosen due to its position as relatively underdeveloped compared to the other two with a lower GDP, lower agri-insurance coverage and generally underdeveloped economic and agricultural infrastructure network, making it a good candidate for a test of the model on countries at earlier stages of development. The 2022 compound hot dry event also impacted Serbian agriculture heavily. Less than 3% of Serbia's total cultivable land has functioning irrigation systems,[9] this makes it almost entirely reliant on consistent rainfall. During this climate event rainfall fell 40-50% below long-term historical averages,[10] creating highly unfavourable conditions for the primary agriculture sector which comprised 5.2% of overall Serbian GDP in 2022.[11] So due to Serbia's comparatively underdeveloped agriculture infrastructure, relatively significant reliance on agriculture and direct common climate impact of historically low rainfall, it was a perfect candidate for applying the model to study the effects of common climate risks such as low rainfall and common infrastructure issues such as weaker insurance systems and less developed capital equipment.

The climate-extended Vasicek model requires five inputs: the unconditional baseline default rate (P^0_D), the probability of a climate hazard occurring (q), the severity of the climate impact (â), a systemic macroeconomic factor (y), the asset correlation factor (ρ). As the theoretical parameters are not directly observable from standard macroeconomic reporting, each variable is mapped onto standardised observable proxies that most closely represent their respective variables consistent across all three countries. The unconditional baseline default rate (P^0_D) is modeled by the 2021 (2021 was chosen as the baseline as it establishes the post pandemic baseline incorporating the acute structural shifts associated with the changes as not to inflate the gap between pre and post climate event NPL rates due to non climate factors) NPL rate and this was done for 2 main reasons: 1) the 90-day delinquency threshold is strictly standardised across all three nations via, 2) the extensive aggregated data is balance sheet and based on realised "defaults" experienced by commercial banks unlike synthetic corporate default models or credit rating agency proxies, aligning much more closely with the theoretical definition given in the ASRF framework. The probability of a climate hazard occurring (q) is modelled by the poisson distribution in this paper; primarily due to its standard industry use in discrete risk hazard modeling and parsimony, more specifically the zero-truncated poisson cumulative distribution function: $P((Hazard) \geq 1) = 1 - P(Hazard\ not\ occurring) = 1 - e^{-\lambda}$. EM-DAT is used for data collection to see how many years from 1980 to 2025 and then summing all the years that either have extreme temperatures or drought or both to align with the events of 2022. This number is then divided by 46 to get the average number of relevant climate events annually which is the value of λ, when substituted into the zero-truncated poisson cumulative formula yields q. The proxy for the severity of the climate impact (â) is determined using the decrease in 2022 agricultural output when compared to 2021 baseline levels for the respective country, this is done by finding the difference in agricultural output from 2021 and dividing by the baseline, $\frac{AGDP\ 2021 - AGDP\ 2022}{AGDP\ 2021}$. This allows the methodology to be standardised across the paper setting 2021 as the baseline and similarly, the standardisation of units and allows for easy international damage comparison while also enabling the shift in standard deviations to be scaled proportionally preserving the Standard normal Gaussian variance. The systemic macroeconomic factor (y) is modeled by finding the year over year growth rate for non agricultural gdp from 2000 to 2022 and then to turn this into a single z-score,

$Z = \frac{YoY\ 2022 - \mu}{\sigma}$ where μ is the average YoY growth rate from 2000-2022 and σ is the standard deviation of the data from 2000 to 2022. This ensures that there is no look-ahead bias, enough raw data points to capture broad market volatility and provide a more accurate Z value and retains the standard normal distribution of the variables which will be needed in the full expression. Asset correlation (ρ) is assigned based on the composition of the most common type of business in the respective domestic agriculture sector according to the Credit risk IRB approach to asset correlation, for example as the vast majority of Spain's farm businesses are classified as corporate (SME corporate if annual sales <$50mm) [12] and thus have an asset correlation between 0.12 and 0.24 and taking the expected value of the range (assuming uniform distribution within the range) outputs 0.18, this process is then repeated for Turkey and Serbia which both receive a value of 0.095 as the asset correlation. The observed NPL rate is taken from q3 2023 reports this was done to ensure that the majority of agricultural defaults had actually occurred as remaining firm reserves run dry as earlier dates could have introduced firms who had been deeply affected but operated on large stockpiles of assets which could be liquidated to maintain loan payments

The code that outputs the end results functions as so: the code uses a data table that stores the 5 inputs as well as the observed NPL rate for Spain, Turkey and Serbia. After defining the climate-extended model by combining expressions 7&8 and weighting them using the probability of occurrence. Each countries variable set is then run through twice, once normally with all inputs as given and then again with everything held constant except â is 0.0 for all 3 datasets. As â approaches 0 the climate-extended model is assumed to collapse back to the base Vasicek model, this allows for a comparison to be made against the general macroeconomic baseline according to the standard Vasicek model. The output statistics of the model are as follows: mean absolute error (MAE) which is used to determine linearly and without any single outsized impact, the magnitude of prediction residuals and root mean square error (RMSE) which is used to emphasise larger deviations and as a result evaluate the models ability to detect tail risk events. While these statistics are overarching to all the whole model, each country also has a set of 5 statistics to quantify results for each nation: Macroeconomic probability of default baseline according to the model when $\hat{a} = 0$, climate-extended probability of default baseline when $\hat{a} > 0$, the marginal climate PD penalty (PD($\hat{a}>0$) - PD($\hat{a}=0$)), the observed NPL rate and the model residual (climate-extended PD - observed NPL). These statistics serve to compare and assess the outputs of the model compared to predicted and observed figures to quantify efficacy of the model under the varying circumstances of each country.

The notable limitations of the model are addressed in this paragraph. Using the poisson distribution for climate hazard probability conflicts with the nature of climate hazards which are becoming more common and thus using historic data to try and calculate the probability of an event occurring that is becoming more and more frequent can potentially underestimate the true value of q. The use of EM-DAT to see the frequency of droughts & severe climate events over the last 46 years means some climate hazards could be missed out on due to specific criteria needed for a climate event to be recorded on EM-DAT. Using 2021 as the baseline while capturing the covid economy and ensuring defaults are not artificially higher also means that natural GDP recovery can lead to artificially stronger macroeconomic conditions rather than just a post-COVID economic recovery. It also doesn't account for the rampant inflation experienced in a lot of these nations that occurred in 2022 and thus could have led to an inflated NPL rate. Using q3 2023 also introduces the risk of missing out defaults before or after this specific 3 month reporting period and could be overstating the time lag component of defaults. The primary root of these issues stem directly from a narrow range of problems. The use of only public data made specific agriculture related inputs that could have potentially aided in developing higher quality methods unfeasible. The inconsistent reporting of a few key agriculture variables directly after covid made certain 2021 and 2022 figures non-standardised across all three countries and thus to maintain consistency, had to switch to less precise data proxies. Data

methodology and collection practices in certain countries especially for long enough periods of time to gather enough data points is a challenging aspect of data collection especially in lower income countries as formatting, reporting terminology and publishing organisations can change all at once from year to year. Basel SME corporate correlation is not a uniform range but rather a deterministic function of PD, this directly affects Spain as the method outlined in the model artificially increases the value of correlation to 0.18 which in an economic upturn suppresses default rates. The proxy for â uses raw percentages which implies 100% volatility, this would result in an underestimation of the climate severity and thus the low marginal climate penalty seen in the model. Finally and most crucially the proxy for probability of defaults is the non performing loan rate. While this may overlap with similar definitions, NPL is a stock of “bad” loans and thus will naturally be higher in almost all cases compared to the actual flow which is what the probability of default measures.

IV - Results

Statistic	**Spain**	**Turkey**	**Serbia**	**Portfolio**
Macroeconomic Baseline PD (â=0)	0.7673%	1.2827%	0.7496%	–
Climate Model PD (â>0)	0.9044%	1.5479%	0.8052%	–
Marginal Climate Penalty	+0.1371%	+0.2652%	+0.0557%	–
Observed NPL	5.2000%	3.8500%	4.9000%	–
Model Residual	-4.2956%	-2.3021%	-4.0948%	–
Mean Absolute Error (MAE)	–	–	–	3.5641%
Root Mean Square Error (RMSE)	–	–	–	3.6751%

Figure 1

Spain had the greatest divergence as the model underestimated risk by 4.2956% as seen in figure 1 the primary cause of this outsized residual likely stems from one specific variable, the macroeconomic Z value. Spain’s post COVID macroeconomic performance which is indicated by the Z score of 1.3197, meaning 1.3197 standard deviations above the mean (as Z is distributed ~ N(0,1)) representing strong outperformance relative to the historical baseline, depress the point-in-time defaults due to abnormally high growth rates being further amplified by Vasicek procyclicality leading to outstandingly low baseline and climate incorporated default rates in the perceived business climate relative to real data. Similarly Spain's rigorous government backed agricultural insurance policy (CAIS) and EU common agricultural policy providing extensive aid and relief for farmers affected by climate events further depress the risk to Spanish farmers,[13] key contributors to Spain's abnormally low predicted default rates relative to the real data.

Turkey's NPL rate prediction and observed NPL rate had the lowest difference at a 2.3021% underestimation. The primary reason for this is due to the rampant inflation in 2023; the unorthodox monetary policy, pre-election spending and exchange rate defence which when dropped led to the Turkish lira losing 37% of its value against the US dollar.[14] This is evident in the Turkish quarterly consumer

inflation reports which in Q3 2023 stood at 61.35%[ref-14.1]. This inflationary pressure on debt is in theory a benefit as the nominal loan burdens fall, however the agricultural credit market in many emerging markets such as Turkey don't rely on fixed contracts.[15] As these short-term and floating rate loans prevent realising the benefits for farmers, the rapid cost push inflation due to spiking imported goods prices e.g fertiliser of which turkey imports 29% of on top of importing roughly 90% of its domestic fertiliser production inputs[ref-15.1]. When these costs outpaced the revenue growth by farmers, it resulted in an elevated default rate. This case also highlights the consequence of using single factor inputs to model broad systemic parameters, as a model can capture climate risk but still may underestimate defaults due to unaccounted exogenous influences. Serbia's case exemplifies the Vasicek procyclicality, with the highest macroeconomic Z value of 1.4989 as a result of the 2021 economic rebound of 9.33% in YoY non-agricultural growth. Although, the non-linear nature of the procyclicality means that the macroeconomic climate does not proportionally scale with Z factor to the degree the Spanish data does. The diminishing marginal gains in the CDF output at the higher end of the standard normal distribution come from the exponential decay of the tails of the distribution, so although the Z value may be higher it is not the sole contributor to the deviation from the observed npl value. Additionally, the â value being 0.0813 and consequent climate penalty being +0.0557% is abnormally low relative to Spain and Turkey (0.1542 & 0.2062 respectively). This is due to the source of the data, as this paper uses the drop in agricultural output due to the 2022 compound hot-dry event and thus ignores other sectors, most notably it ignores logistics. Inland water transport accounts for approximately 4.7% of Serbia's total freight transport,[16] however this changes in key trade industries such as fuel, construction and grain. The Danube can make up a significant proportion of inland water imports and exports, for example about 60-70% of Serbian grain exports occur via river barges.[17] So when extreme temperature and prolonged drought occur the Danube can drop to critical levels affecting these key industries. This is evident in the fact that when the Danube drops to these levels, tankers imports of fuel can drop to as low as 25% of planned targets.[18] The hidden cost of climate knock-on effects in these vital industries with countless secondary consequences is lost when isolating agricultural data and a substantial reason for the underestimate. These main structural contributors are also compounded by the generally slower financial and legal institutional processes within developing countries particularly when it comes to illiquid secondary markets for distressed debt causing NPL defaults to stay on balance sheets long after the shock or the default that caused it has passed. All these factors lead to both an overinflated observed NPL rate and an underestimated climate and macroeconomic risk aiding in the noticeable 4.0948% underestimate. To evaluate the models performance holistically across all three countries, the three nations were treated as being in a single portfolio to then which MAE and RMSE data were outputted. As seen in figure 1, MAE was recorded to be 3.5641% and RMSE was recorded to be 3.6751%. MAE being at 3.5641% in absolute terms is a physically small value but becomes immediately significant in context of NPL default rates which in this case varied from 3.85% to 5.20% meaning the MAE was a majority of all NPL values observed and multiples of the predicted. However the close spread of 11.1 basis points between the MAE and RMSE indicates that the error is likely systematic and near-uniform rather than driven by extreme anomalous results, likely originating from structural factors in the input data than the model. It also highlights the model's stability under drastic regime changes. For example, strong economic climates in Spain and Serbia during 2021 suppressed default rate predictions but the slightly more extreme systemic input of Serbia being roughly 0.2 higher did not cause severe tail end instability by disproportionately scaling down the probability of default. Overall the model behaves well in general, having mostly systematic errors due to single factor inputs that fail to capture specialised domestic structural flaws.

V - Conclusion

The integration of physical risk parameters into the standard Vasicek ASRF framework has outlined two primary mechanisms that govern climate-adjusted credit risk. First, the fact that macroeconomic climates have an outsized impact on overall predicted default rates. As the macroeconomic variable Z follows the standard normal distribution, extreme economic environments such as recessions or post-recession rebounds can heavily distort perceived economic performance relative to historical averages. This theory applies to an extent as after a Z value of ± 1 (standard normal inflection points) the marginal cumulative probability gained by increasing the magnitude of the Z value decays exponentially and thus increasing or decreasing this value at either extreme won't affect the model proportionately, an aspect especially useful when dealing with developing nation climate-credit risk modeling as volatile economic conditions are often consistently unstable. Secondly, single factor climate inputs based on a single measure are unreliable as the basis for a global framework. For developed countries such as Spain, climate risk especially in the agricultural sector manifests directly via destroyed crop yields. This is frequently untrue for developing nations as farming is often decentralised and run on a much smaller scale often as a family enterprise, so when disasters do occur they often affect a localised group and this is commonly insufficient to affect aggregate national data. Rather than directly affecting crop yields, logistical or infrastructure vulnerabilities are more likely to be the major contributors to the severity of climate impacts. As a result, relying solely on single factor inputs creates empirical blind spots in emerging markets, requiring a multi-factor extension or composite single factor input to adjust for these situations.

Beyond theoretical insights, the climate-extended Vasicek model provides direct operational utility in quantitative risk analysis and regulatory compliance within credit markets. Climate risk parameter such as climate severity and hazard probability (â & q) can not only be readily implemented in the climate-extended Vasicek model, but also to calibrate forward looking macroeconomic condition predictions. This is particularly relevant due to forward looking macroeconomic conditions being a legal requirement when staging stage 1 and stage 2 loans under IFRS 9. The simple and relatively unspecialised variables mean direct and broad applications can be made to further the accuracy of these outlooks providing a more realistic point of view to institutional risk management. As the model is based directly on the Vasicek model and uses standardised variables and distributions, it offers seamless integration with current Basel frameworks and presents an easily interpretable and formulaic gaussian model for regulators and risk committees to defend when assessing adequate capital buffers. Although this model offers many benefits relative to standard credit risk modelling, the single factor inputs often create numerous blind spots which can lead to sizable unaccounted risks for certain positions. Switching to a through-the-cycle output and applying post-model adjustments are necessary to equip these models with the dynamic responsiveness needed for more complex risk modelling.

As seen previously this model exhibits a few major limitations in the accuracy of determining credit default probabilities. Future iterations could improve on these drawbacks through these refinements: transition from a single year baseline to a multi-year rolling average (or applying a Hodrick Prescott filter) to exclude transient short term economic shocks when recording baseline and observed data. Enhancing the climate hazard severity and probability figures by relying less on broad gdp figures, averages and EM-DAT threshold style climate reporting and more on World Meteorological Organisation data, remote crop yield indices, logistical and other such sector related data.

Ultimately, while single factor ASRF models face structural issues such as procyclicality compression, they provide the ultimate convenience of transparent, computationally efficient and easily integratable blueprints for incorporating climate risk directly into the systematic risk factor rather than arbitrary ad-hoc overlays. This renders the fundamental idea behind the process invaluable for future credit risk analysis and stress testing - an imperative as climate change intensifies and continues to impact financial markets across the globe.

References

[1] EM-DAT, *Emdat.be*, 2020. https://public.emdat.be/data (accessed Sept. 20, 2026). (2000-2026, drought, africa,europe)

[2] E. waters:, “Europe’s waters: key challenges and EU solutions | Topics | European Parliament,” *Topics | European Parliament*, Feb. 05, 2025. https://www.europarl.europa.eu/topics/en/article/20250428STO28142/europe-s-waters-key-challenges-and-eu-solutions (accessed Sept. 20, 2026).

[3] “Water savings for a water-resilient Europe,” *Europa.eu*, June 04, 2025. https://www.eea.europa.eu/en/analysis/publications/water-savings-for-a-water-resilient-europe (accessed Sept. 20, 2026).

[4] ESRB, “Climate-related risks and accounting,” 2024. Accessed: Sept. 20, 2026. [Online]. Available: https://www.esrb.europa.eu/pub/pdf/reports/esrb.report202404_climaterelatedrisks~2311dfaee2.en.pdf

[5] Spain's, “At a glance.” Accessed: Sept. 20, 2026. [Online]. Available: https://agriculture.ec.europa.eu/document/download/337fcbc6-19a2-4c91-945b-248d63a0134c_en?filename=csp-a-a-glance-spain_en.pdf

[6] “ITALY’S CAP STRATEGIC PLAN.” Accessed: Sept. 20, 2026. [Online]. Available: https://agriculture.ec.europa.eu/document/download/34058aa4-eee5-4579-99af-0e7c100d40ba_en?filename=csp-at-a-glance-italy_en.pdf

[7] E. Metreau, K. E. Young, and S. G. Eapen, “Who moves up and why? A closer look at the new World Bank Group Country Income Classifications Release,” *World Bank Blogs*, July 03, 2026. https://blogs.worldbank.org/en/opendata/who-moves-up-and-why--a-closer-look-at-the-new-world-bank-group- (accessed Sept. 20, 2026).

[8] “What Türkiye’s insurance pool shows about risk sharing at scale,” *undp_sdgfinance*, 2026. https://irff.undp.org/blog/what-turkiyes-insurance-pool-shows-about-risk-sharing-scale (accessed Sept. 20, 2026).

[9] “Serbia eyes sustainable irrigation,” *Agroberichtenbuitenland.nl*, June 20, 2025. https://www.agroberichtenbuitenland.nl/actueel/nieuws/2025/06/20/serbia-sustainable-irrigatiano (accessed Sept. 20, 2026).

[10] Branko Petrović, “Serbia’s Drought Halves Crop Yields, Few Farms Insured,” *Srpske Novine*, Aug. 29, 2026. https://srpske.rs/en/news/ekonomija/2026/08/29/serbia-drought-halves-yields-few-farms-insured (accessed Sept. 20, 2026).

[11] T. Novaković, M. Tomaš Simin, D. Novaković, and B. Mutavdžić, “ANALYSIS OF THE GROSS ADDED VALUE OF AGRICULTURAL PRODUCTION IN THE REPUBLIC OF SERBIA,” *Ekonomika poljoprivrede*, vol. 71, no. 4, pp. 1383–1399, Dec. 2024, doi: 10.59267/ekopolj24041383n.

[12] “Basel Committee on Banking Supervision The Basel Framework.” Accessed: Sept. 21, 2026. [Online]. Available: https://www.bis.org/committees/bcbs/basel-framework/basel-framework.pdf

[13] Spanish multi-peril agricultural insurance scheme: linchpins of crop and livestock farming in Spain,” *Number*, vol. 22, 2025, Accessed: Sept. 20, 2026. [Online]. Available: https://consorsegurosdigital.com/almacen/pdf/x_ray_image.pdf

[14] N. Turak, “Turkish lira hits fresh record low against the U.S. dollar,” *CNBC*, Jan. 11, 2024. https://www.cnbc.com/2024/01/11/turkish-lira-hits-fresh-record-low-against-the-dollar.html (accessed Sept. 21, 2026).

[15] “INFLATION REPORT 2023-IV,” 2023. Accessed: Sept. 21, 2026. [Online]. Available: https://www.tcmb.gov.tr/wps/wcm/connect/bc7b5b97-2760-4824-82c8-afda35ec5f81/Inflation+Report+2023iv.pdf?MOD=AJPERES&CACHEID=ROOTWORKSPACE-bc7b5b97-2760-4824-82c8-afda35ec5f81-oOwZWeD (ref14.1)

[16] U. Gültekin, “Factors determining farmers’ access to agricultural credit in Turkey,” *Ciência Rural*, vol. 53, no. 9, 2023, doi: 10.1590/0103-8478cr20210916. (15)

[17] Turkish Minute, “Turkey seeks Russian fertilizer deal as Gulf supply disruptions hit imports: report,” *Turkish Minute*, Sept. 08, 2026. https://www.turkishminute.com/2026/09/08/turkey-seeks-russian-fertilizer-deal-as-gulf-supply-disruptions-hit-imports-report/ (accessed Sept. 21, 2026). (ref15.1 A)

[18] “Hürriyet Daily News,” *Hürriyet Daily News*, Apr. 07, 2026. https://www.hurriyetdailynews.com/war-puts-turkish-farmers-in-a-low-yield-or-high-cost-dilemma-220721 (accessed Sept. 21, 2026). (ref15.1 B)

[19] Milanković Jovanov, Jelena & Pavic, Dragoslav & Đorđević, Jasmina & Dragin, Aleksandra & Đukičin, Smiljana & Mészáros, Minucsér. (2018). The Danube inland waterway transport and its role in Serbia's economic development. R-Economy. 4. 105-114. 10.15826/recon.2018.4.3.015. (16)

[20] N. D. R. Ajansı, "'We fight for our place at the grain market with good quality and competitive prices,'" *Millermagazine.com*, 2026. https://millermagazine.com/blog/we-fight-for-our-place-at-the-grain-market-with-good-quality-and-competitive-prices-3407 (accessed Sept. 21, 2026). (17)

[21] A. TORETI *et al.*, "Drought in Europe August 2022," *JRC Publications Repository*, Dec. 2022, doi: 10.2760/264241. (18)

[22] S. Marques -Production and C. Director, "X-ray image of Agroseguro and the Spanish multi-peril agricultural insurance scheme: linchpins of crop and livestock farming in Spain X-ray image of Agroseguro

[23] "Serbia Archives," *FocusEconomics*, Feb. 08, 2024. https://www.focus-economics.com/countries/serbia/ (accessed Sept. 20, 2026).

[24] "Financial needs in the agriculture and agri-food sectors in Spain," 2020. Accessed: Sept. 20, 2026. [Online]. Available: https://www.fi-compass.eu/sites/default/files/publications/financial_needs_agriculture_agrifood_sectors_Spain.pdf

[25] "Base de Datos para el Control Integral de acceso a Subvenciones (CIAS)," *Ministerio de Agricultura, Pesca y Alimentación*, 2022. https://www.mapa.gob.es/es/enesa/asegurados_solicitud_obtencion_subv (accessed Sept. 20, 2026).

[26] "Informes de Financiación en los Sectores Agrario, Pesquero y Agroalimentario en España," *Ministerio de Agricultura, Pesca y Alimentacion*, 2021, 2023. https://www.mapa.gob.es/es/ministerio/servicios/analisis-y-prospectiva/financiacion-fiscalidad-costes-energia/informes-financiacion-endeudamiento (accessed Sept. 21, 2026).

[27] "CONTENTS." Accessed: Sept. 21, 2026. [Online]. Available: https://www.ziraatbank.com.tr/en/Investor-Relations-ZB/Financials/Documents/ziraat_bankasi_entegre_faaliyet_raporu_2021-eng.pdf

[28] "Annual Financial Stability Report - 2021," Financial Stability Department, Belgrade, Serbia, Jun. 2022. [online]. Available: nbs.rs Accessed: Sept. 21, 2026.

[29] EM-DAT, *Emdat.be*, 2020. https://public.emdat.be/data (accessed Sept. 20, 2026). (1980-2026, drought, extreme temperatures,turkey,serbia,spain)

[30] Eurostat, “Agriculture output: 19% value rise fuelled by price surge,” *@EU_Eurostat*, Nov. 16, 2023. https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20231116-2 (accessed Sept. 21, 2026).

[31] “Agriculture and Rural Development Statistical Factsheet,” 2021. Accessed: Sept. 21, 2026. [Online]. Available: https://agriculture.ec.europa.eu/system/files/2021-12/agri-statistical-factsheet-es_en_0.pdf

[32] Globalen LLC, “Turkey Agriculture value added - data, chart | TheGlobalEconomy.com,” *TheGlobalEconomy.com*, 2025. https://www.theglobaleconomy.com/Turkey/value_added_agriculture_dollars/ (accessed Sept. 21, 2026).

[33] T. Novaković, M. Tomaš Simin, D. Novaković, and B. Mutavdžić, “ANALYSIS OF THE GROSS ADDED VALUE OF AGRICULTURAL PRODUCTION IN THE REPUBLIC OF SERBIA,” *Ekonomika poljoprivrede*, vol. 71, no. 4, pp. 1383–1399, Dec. 2024, doi: 10.59267/ekopolj24041383n.

[34] “Statistical Release | Statistical Office of the Republic of Serbia,” *www.stat.gov.rs*. https://www.stat.gov.rs/en-us/vesti/statisticalrelease/?p=13820 (accessed Sept. 21, 2026).

[35] “Annual Report 2023,” *Annual Report*, Apr. 2024, doi: 10.53479/36513.

[36] “Veri Detay,” *Bddk.org.tr*, 2025. https://www.bddk.org.tr/Veri/Detay/171 (accessed Sept. 21, 2026). Turkish data

[37] “Annual Financial Stability Report 2023,” *nbs*, 2023. https://www.nbs.rs/en/drugi-nivo-navigacije/publikacije-i-istrazivanja/GISFS/ (accessed Sept. 21, 2026).

[38] World Bank, “GDP (current US$) - Spain | Data,” *data.worldbank.org*, 2023. https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=ES (accessed Sept. 21, 2026).

[39] The World Bank, “GDP (current US$) - Turkey | Data,” *Worldbank.org*, 2023. https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=TR (accessed Sept. 21, 2026).

[40] “GDP (current US$) - Serbia | Data,” *data.worldbank.org*. https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=RS (accessed Sept. 21, 2026).

[41] “Spain - Agriculture, Value Added (% Of GDP) - 2023 Data 2024 Forecast 1995-2021 Historical,” *tradingeconomics.com*. https://tradingeconomics.com/spain/agriculture-value-added-percent-of-gdp-wb-data.html (accessed Sept. 21, 2026).

[42] “Turkey GDP From Agriculture,” *Tradingeconomics.com*, 2025. https://tradingeconomics.com/turkey/gdp-from-agriculture (accessed Sept. 21, 2026).

[43] “Serbia GDP From Agriculture,” *tradingeconomics.com*. https://tradingeconomics.com/serbia/gdp-from-agriculture (accessed Sept. 21, 2026).

[44] 0. A. Vasicek, “Probability of loss on a loan portfolio,” KMV Corporation, San Francisco, CA,USA, Tech. Rep., 1987

[45] R. C. Merton, “On the Pricing of Corporate Debt: the Risk Structure of Interest Rates,” *The Journal of Finance*, vol. 29, no. 2, pp. 449–470, May 1974.

[46] M. B. Gordy, “A risk-factor model foundation for ratings-based bank capital rules,” *Journal of Financial Intermediation*, vol. 12, no. 3, pp. 199–232, Feb. 2003, Accessed: Sept. 21, 2026. [Online]. Available: https://ideas.repec.org/a/eee/jfinin/v12y2003i3p199-232.html#article

[47] V. Singh, R. Ordonez, and J. Law, “Addressing Physical Risk Impacts in Risk Management Practices,” Moody’s, Dec. 2025.

[48] V. Pozdyshev, A. Lobanov, and K. Ilinsky, “Incorporating physical climate risks into banks’ credit risk models,” 2025. Accessed: Sept. 21, 2026. [Online]. Available: https://www.bis.org/publications/working-paper-1274-incorporating-physical-climate-risks-banks-credit-risk-models.pdf

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Appendix A: Code

```
import numpy as np
from scipy.stats import norm
def climate_vasicek(pd0, q, â, y, rho=0.15):
    CDT = norm.ppf(pd0)
    IRD = np.sqrt(1 - rho)
    expression_7 = (CDT - np.sqrt(rho) * y) / IRD
    expression_8 = (CDT - np.sqrt(rho) * y + â) / IRD
return (1 - q) * norm.cdf(expression_7) + q * norm.cdf(expression_8)
countries = {
    'Spain': {
        'pd0': 0.051,
        'q': 0.3095,
        'â': 0.1542,
        'y': 1.31966003382962,
        'rho': 0.18,
        'npl_obs': 0.052
    },
    'Turkey': {
        'pd0': 0.0252,
        'q': 0.2901,
        'â': 0.2062,
        'y': 0.539292388103927,
        'rho': 0.095,
        'npl_obs': 0.0385
    },
    'Serbia': {
        'pd0': 0.032,
        'q': 0.2835,
        'â': 0.0813,
        'y': 1.49886241679232,
        'rho': 0.095,
        'npl_obs': 0.049
    }
}
```

```
errors_climate = [
    climate_vasicek(
        pd0=p['pd0'],
        q=p['q'],
        â=p['â'],
        y=p['y'],
        rho=p['rho']
    ) - p['npl_obs']
    for p in countries.values()
]
macro_pd_values = [
    climate_vasicek(
        pd0=p['pd0'],
        q=p['q'],
        â=0.0,
        y=p['y'],
        rho=p['rho']
    )
    for p in countries.values()
]
mae_climate = np.mean(np.abs(errors_climate))
rmse_climate = np.sqrt(np.mean(np.square(errors_climate)))

print(f"Climate Model MAE:  {mae_climate:.4%}")
print(f"Climate Model RMSE: {rmse_climate:.4%}")
for country_name, p in countries.items():
    climate_pd = climate_vasicek(
        pd0=p['pd0'], q=p['q'], â=p['â'], y=p['y'], rho=p['rho']
    )
    macro_pd = climate_vasicek(
        pd0=p['pd0'], q=p['q'], â=0.0, y=p['y'], rho=p['rho']
    )
    climate_penalty = climate_pd - macro_pd
    model_residual  = climate_pd - p['npl_obs']
    print(f"--- {country_name} ---")
    print(f"Macro Baseline PD (â=0):   {macro_pd:.4%}")
```

```
print(f"Climate Model PD (â>0):    {climate_pd:.4%}")
print(f"Marginal Climate Penalty:  {climate_penalty:+.4%}")
print(f"Observed Target NPL:       {p['npl_obs']:.4%}")
print(f"Model Residual:     {model_residual:+.4%}\n")
```

Appendix B: Input Data

	Spanish non agriculture gdp		Turkish non agriculture gdp		Serbia non agriculture gdp	
Date	Real GDP in Trillions usd	YoY NA GDP growth rates	Real GDP in Trillions usd	YoY NA GDP growth rates	Real GDP in Billions usd	YoY NA GDP growth rates
2000	0.95503	2.736048262	274.72248	7.044523566	19.48	10.30577576
2001	0.99605	4.295153032	202.18227	-26.40490505	20.52	5.338809035
2002	1.01414	1.816173887	240.75552	19.07845332	22.11	7.748538012
2003	1.03818	2.370481393	315.35918	30.98731028	23.42	5.92492085
2004	1.07581	3.624612302	410.12186	30.04912684	24.68	5.380017079
2005	1.11692	3.821306736	508.26359	23.92989489	26.27	6.442463533
2006	1.15669	3.56068474	559.62733	10.1057288	27.43	4.415683289
2007	1.20136	3.861881749	685.17898	22.4348675	30.03	9.478672986
2008	1.21251	0.9281148032	775.3628	13.16208212	31.47	4.795204795
2009	1.16686	-3.764917403	653.83759	-15.67333511	30.36	-3.527168732
2010	1.16655	-0.02656702604	782.48033	19.6750297	30.85	1.613965744
2011	1.15326	-1.139256783	844.12144	7.877656171	30.84	-0.03241491086
2012	1.12712	-2.266618109	885.26224	4.873801097	31.21	1.199740597
2013	1.10322	-2.120448577	962.10606	8.680345386	30.85	-1.15347645
2014	1.12272	1.767553163	942.27843	-2.060856991	30.19	-2.139384117
2015	1.17819	4.940679778	865.40064	-8.158712707	30.51	1.059953627
2016	1.20636	2.390955618	870.76658	0.6200526961	31.26	2.458210423
2017	1.2434	3.070393581	863.81848	-0.7979291075	32.45	3.80678183
2018	1.27081	2.20443944	788.31455	-8.740717147	33.64	3.667180277
2019	1.29536	1.931838749	775.80109	-1.5873689	35.46	5.410225922
2020	1.15407	-10.90739254	733.57781	-5.442539401	34.93	-1.49464185
2021	1.23052	6.624381537	839.89534	14.49301336	38.19	9.332951618
2022	1.31567	6.919838767	926.05747	10.25867461	39.58	3.639696256
μ		1.593014657		6.713225909		3.637900242
σ		3.812623518		14.42591741		3.799582478
Z-value		1.319660034		0.5392923881		1.498862417

Parameter	Spain	turkey	Serbia
Baseline Default Rate (P^{0}_{D})	0.051	0.0252	0.032
Disaster probability (q)	0.3095	0.2901	0.2835
Damage Severity (â)	0.1542	0.2062	0.0813
Macroeconomic factor (Z)	1.319660034	0.5392923881	1.498862417
Asset correlation (ρ)	0.18	0.095	0.095
Observed Post NPL (P^{obs}_{D})	0.052	0.0385	0.049